

Homework is Dead, Long Live Homework

Can we improve OSU's undergraduate homework system?



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Introduction

The aim of this paper is to spark discussion about the workload in OSU's undergraduate math courses. We and other members of the department believe that students in MTH 112Z or MTH 251-256 are required to complete too much work, so we propose several potential alternatives to the current system. In writing the paper, we have tried to abide by two principles. We ask the same of the reader:

- Try to be open-minded. We understand that you might disagree with our assessment of the current system and the alternatives, and we are very interested in your feedback. But please try not to dismiss an alternative just on face-value; we're all trying to find *good reasons* while ignoring bad ones.
- Consider a variety of perspectives. Please make sure that you take into account a wide range of perspectives—not only students of various socio-economic backgrounds, but also GTAs, instructors, administrators, and course coordinators.

We don't expect the reader will have any trouble with these :)

In the next section, we'll closely examine the pros and cons of the current system. Then, we'll present a radical alternative and several compromises. At the end is a discussion of some important related topics.

We are presenting alternatives for MTH 112Z or MTH 251-256. However, for clarity and brevity, we'll restrict our discussion to MTH 251.

The current system

Figure 1 shows the workload for MTH 251 as taken from a syllabus.

ONLINE HOMEWORK (ACHIEVE) – DUE THURSDAYS & SUNDAYS BY 11:59PM

The online homework is a time to practice solving problems and applying conceptual ideas mathematically. The online homework system will tell you immediately if you have “gotten the correct answer” or not. This allows you to self-correct and learn from your mistakes as you work the problems.

WRITTEN HOMEWORK & REFLECTIONS – DUE THURSDAYS & SUNDAYS BY 11:59PM PDT

Written Homework – Due Thursdays

The written homework is a great way to practice problems without the added structure and guidance that the online homework system provides. This should help you (the student) to assess what your strengths and weaknesses are. Attempting to solve these problems is an important part of understanding what we are learning.

Written Homework Reflection – Due Sundays

By looking at the provided solutions and comparing them to your own solutions you will start to find what your weaknesses and strengths are in the material covered. By responding about why you made mistakes or had misconceptions about a problem(s), you are solidifying that understanding while practicing your communication skills!

COURSE ENGAGEMENT

Pre-Class Quizzes – Due each day before class

The pre-class quizzes provide an opportunity to practice skills introduced in the pre-class materials (videos and readings), answer survey questions, and inform your instructor about questions you will have about the pre-class materials. These quizzes will be graded on correctness. *Your three lowest pre-class quiz scores will be dropped.*

Additional Activities / In Class Activities

Some days we will have additional problems to work on that need to be submitted to Canvas or Gradescope for course credit.

Recitation Activities – Done in Recitation

We will be working on active learning activities during each weekly recitation. These activities could be answering poll questions, working on a group activity with your peers, reflecting on the recent big ideas in the course, or other active learning activities. You will earn points for these activities by being an active participant and answering questions posed in recitation. *Your lowest Recitation score will be dropped.*

UNIT MINI-EXAMS – DURING WEEKS 3, 5, 7, 9, & FINALS WEEK

Unit quizzes will assess procedural fluency and concepts based on key learning outcomes.

Figure 1: The MTH 251 workload

If you're keeping count, that's about 10 assignments per week and an exam every other week. When calculating a student's final grade, these assignments are weighted according to the grade breakdown in Figure 2.

ASSESSMENT & GRADING	
CATEGORY	PERCENT
Online Homework – Achieve	14%
Written Homework and Reflections	14%
Course Engagement	12%
Unit 1 Mini-Exam	12%
Unit 2 Mini-Exam	12%
Unit 3 Mini-Exam	12%
Unit 4 Mini-Exam	12%
Unit 5 Mini-Exam	12%
Total	100%

Figure 2: The MTH 251 grade breakdown

In total, homework assignments compose 40% of a student's final exam grade, while exams compose the remaining 60%.

Pros of the current system

Lots of practice

It is undeniable that the current system embodies a commitment to practice. Since students are assigned approximately 10 homework assignments per week, with most assignments containing more than 8 problems, they spend most of their non-class time doing problems. In many ways, this makes a lot of sense. It's hard to over-state just how important practice is for learning mathematics, so this is clearly one the current system's areas of strength.

Feedback

Some assignments in the current system provide students with more than just an opportunity to practice. The Achieve problems, some written homework problems, the pre-class quizzes, and the written homework reflections all give feedback on how the student is doing. This kind of formative assessment is useful both for the students and for the instructor as a way of gauging the student's understanding in the middle of the learning process. So, the amount of feedback is definitely a part of the system we don't want to change.

Summative assessment matters most

Despite the amount of formative assessment in the homework categories, they constitute only 40% the student's final grade. The remaining 60% of the student's final grade comes

from written exams. Written exams are a kind of summative assessment: they are used to evaluate the student's understanding and content mastery at the end of the learning process. In the next section, we'll argue that summative assessment should make up *even more* of the student's final grade, but let's keep in mind that the current system already succeeds in this area to some extent by having summative assessments play a crucial role in student success.

Cons of the current system

In this section, we argue that the current system is a major threat to the general learning quality of a student. In particular, we believe that the current system results in more students who pass when they shouldn't, less student autonomy, more cheating, and lower motivation and appreciation for math. We support these conclusions with a framework for understanding how students approach different learning settings in different ways.

Homework is not summative assessment

Many people see a student's final grade as a summative assessment of their knowledge: an A+ student understands math better than a C- student.¹ But since this is the case, it is odd for homework to contribute to the final grade. After all, homework—especially pre-class quizzes where students have just begun to engage with the content—occurs in the *middle* of the learning process. At this point, students are *expected* and *encouraged* to make mistakes, so the question is this: if we tell students that homework mistakes are a normal, useful part of the learning process, why do they negatively contribute to our assessment of the student's knowledge?

Students are not homogeneous

When students are graded on 10 homework assignments per week, this leaves them with no time to learn in ways that *they* choose. In the middle of the term, some students might benefit most from watching someone else solve problems, while others might benefit most from a review of pre-requisite content. However, the current system discourages students from engaging in this way: they are required to complete 40 Achieve problems, so that is what they will prioritize whether it will help them learn or not. In other words, the current system has students do the same learning practices, as if every student had the same needs and skills.

School is not work

In [1], Desforges argues that students do not simply absorb knowledge in a particular context, but rather adapt to each classroom in a multi-faceted way (81). He describes how this often results in the development of a *school-as-work* approach:

¹There is debate about whether final grades *should* be seen this way, but we ignore these concerns for now.

There is evidence that most children adapt quickly to school as a job. They organize their resources—intellectual, social, and material—to do their work. [...] The [school-as-work approach] is highly adapted to identifying the procedural needs of the school tasks and to delivering a product that meets the teacher's assessment criteria. (81-82)

In other words, the school-as-work approach is a mindset in which the student cares more about getting the work done than they do about learning the content.

In our experience, many math students at OSU take a school-as-work approach, and we think that the workload is a major reason why. 10 assignments per week, including more than 40 Achieve problems, is simply a staggering amount for a student in 2 (or more) other courses; imagine if they are a parent or working a part-time job! When a student is faced with such a large amount of work, it is likely that they will see the practical challenge of completing the work—not the educational challenge of learning the content—as their biggest challenge and their ultimate goal.

The main problem with the school-as-work approach is that it is deceiving. Presumably, most students are at an educational institution to get *educated* and learn. But this approach does not have students aim at learning; it has them aim at checking off their to-do list of assignments. Desforges writes about how this manifests when students encounter hurdles in the learning process:

This is not perceived as learning difficulty at all but rather work that is too hard. The response is to pressure the teacher to reduce the workload or work challenge. If students in this case try harder, they do not try harder to learn. Rather, they will try harder to complete the work or try harder to have the challenge in the work diminished. From this theoretical perspective, many if not most students do not see themselves as learners at all. (82)

We think that OSU students often aim at the wrong goal in this way, and that this hurts the things that matter: learning math, appreciating math, and appreciating problem-solving and learning in general. Let's briefly explore these negative consequences further.

Encouraging bad behavior

One side-effect of students seeing school as work is *cheating*, such as submitting work done by Chat-GPT or other AI platforms like Mathway. Cheating is bad because a grade should reflect a student's understanding and cheating thwarts that goal, resulting in students who pass the class without actually understanding the material. Every graduate student I know has an experience with such a student whose prior mathematical knowledge is not sufficient enough to understand new concepts and succeed in their current math course. Cheating also builds a habit of academic dishonesty, which hurts our broader society through misinformation, negligence, and ignorance. But if a student's ultimate goal is producing satisfactory work,

using online AI platforms stops looking like a bad habit and starts looking more like a tool for success.

Below is a table which displays what percentage students must earn in the exam portion of the class given the non-exam portion grade and the desired course grade. We have grouped the written homework, achieve, and course engagement categories to represent the non-exam grade (40% of the overall course grade) and grouped the 5 exams into the required exam grade (60% of the course overall grade). For example, if a student receives 90% on the non-exam portion of the course, and they desire a 75% in the class, the student only needs to score a 65% on their exams. This grading structure incentivizes students to successfully complete all the assignments in the non-exam portion of the class, allowing sub-par understanding during summative assessments.

		Desired course grade						
		100	95	90	85	80	75	70
Non-Exam Grade	100	100	91.7	83.3	75	66.7	58.3	50
	95	103.3	95	86.7	78.3	70	61.7	53.3
	90	106.7	98.3	90	81.7	73.3	65	56.7
	85	110	101.7	93.3	85	76.7	68.3	60
	80	113.3	105	96.7	88.3	80	71.7	63.3
	75	116.7	108.3	100	91.7	83.3	75	66.7
	70	120	111.7	103.3	95	86.7	78.3	70

Table 1: Required exam grade by desired course grade and current non-exam grade

Motivation and appreciation

The school-as-work approach also hampers student motivation and appreciation of mathematics. Mathematics is useful, mathematics is beautiful, and mathematics disciplines the mind, but students rarely see it that way. Students who see school as work are less likely to take time to think about the beauty or usefulness of a mathematical concept. They see learning difficulties as barriers to success, not transformative experiences which give rise to mental discipline. This is exacerbated by punishing students for making useful mistakes.

The end-result is that many students do not appreciate mathematics. Challenges prompt frustration, self-doubt, and low confidence, not motivation to learn.

Autonomy and learning skills

In the school-as-work approach, students become accustomed to completing whatever is assigned to them. But when students graduate and enter the real world, learning hardly occurs in such a structured environment. As such, the school-as-work approach does not prepare students for the learning process; instead, it trains them to mechanically complete any work given to them. Compared to these automatic task-completion skills, autonomous learning skills are more valuable and closer to the goals of education, so the current system has lots of room for improvement in this area.

The radical alternative

In [2], Joe Feldman discusses some of the cons presented in the previous section and comes to a surprising conclusion: homework should not be included in the grade. Instead, he argues, we should abolish homework grades and base 100% of a student's grade on their summative assessment. This doesn't mean getting rid of practice or feedback in general; only that students would not be *graded* on practice. We call such a system the *radical alternative*. This section is dedicated to discussing some of this alternative's pros and cons. In the next section, we'll explore some possible compromises between the current system and the radical alternative.

Pros of the radical alternative

Good mistakes go unpunished

In the current system, homework mistakes lead to lower course grades, even though these mistakes are useful parts of the learning process. If we were to adopt the radical alternative and abolish homework grades, this would no longer be an issue. Mistakes would finally take their rightful place as a good, healthy part of learning, and course grades would more accurately reflect the student's understanding at the end of the learning process.

Flexibility

Under the current system, all students are required to complete homework assignments even when that might not be what's best for their learning. The radical alternative would not run into this issue, as students would have no incentive to complete assignments that were unhelpful for their learning. If a student wanted to watch lecture videos or see someone else solve problems, they could do that. Students would no longer have to fear punishment for learning in un-assigned or atypical ways, giving them more flexibility to learn in the way that is best for them.

Learning over working

The current system encourages students to adapt to school as a job instead of as a learning experience. Students who see school as work prioritize doing their work instead of engaging with the content and learning the material. If we got rid of homework grades, students would be incentivized to learn instead of being incentivized to do work. This has many benefits, one of which is that it helps students practice the process of learning. It must be asked: Are our math classes giving students the skills to learn other things once they graduate? Under the radical alternative, this question is easily answered: if a student doesn't practice learning in their calculus class, they won't succeed. In other words, passing calculus would necessitate the practice of learning.

Goodbye cheating, goodbye AI

Homework grades exist in part to motivate students to practice. I hear you say, "If students don't get graded on homework, they won't do it!" But this misleadingly implies that the current system forces students to practice. The current system incentivizes cheating and AI-use to complete their work without engaging with the material, and we have seen this happen first-hand. So, even in the current system, many students who do not want to do the work are already choosing not to do it. Under the radical alternative, there would be no incentive to do this. If we stop rewarding students for doing work and only reward them for understanding the material, cheating and AI no longer arise as issues.

Intrinsic motivation

The claim that "If students don't get graded on homework, they won't do it!" is misguided for another reason: it implies that the extrinsic motivator of "earning points" is the only reason students will engage. However, in [2], Feldman writes

we know from a century of research after Skinner that though extrinsic punishments and rewards may be effective when it comes to incentivizing menial, repetitive tasks like stuffing envelopes, extrinsic motivation fails when tasks require creativity and complex thinking. Because most of us consider homework to be more intellectually demanding than stuffing envelopes and to require complex and creative thinking (at least we intend it to be), extrinsic motivation strategies are likely to fail. To motivate someone to successfully perform intellectually demanding tasks, intrinsic motivation is much more powerful and effective. (138)

Grades provide extrinsic motivation for students to work, but the work is too intellectually demanding for this to be a valid approach. This provides some explanation for cheating and AI-use, and it highlights another benefit of the radical alternative: if we abolish homework grades, students will have no extrinsic motivation to practice. Instead, if they are motivated

to do the work at all,² they will be motivated only by their desire to learn, understand, and succeed on large-scale assessment.

Play

In [3], Francis Su writes

Doing math properly is engaging in a kind of play: having fun with ideas that emerge when you explore patterns, and cultivating wonder about how things work. Math is not about memorizing procedures or formulas, or at least that's not where you start. It's the same way in sports. In football, you wouldn't practice drills unless you wanted to play competitively, but you can start with an enjoyment of the game. (50)

The current system does not give students time to play or explore ideas; the workload is just too big, and all of their work is pushing them towards efficiency in producing a solution, not playing with the ideas. Under the radical alternative, students would have more time to engage in these practices, which would increase students' appreciation of the mathematics they are learning.³ We could provide resources which discuss the content conceptually and holistically, and we could provide problems which provoke exploration and mystery.

Cons of the radical alternative

Change requires effort

The most immediate and practical downside to adopting the radical alternative is that it would take time and effort. This is an unavoidable part of all large-scale systemic change, and it is a definite con because continuing with the current system would be easy. Also, the math department's faculty and staff already do an enormous amount of work, so any significant increase in their responsibilities would make changing to the radical alternative essentially impossible.

But we should clarify just how much of a downside this is. First, just because change requires time and effort does not mean that change is not worth pursuing. Because time and effort are unavoidable in all change, if we view them as impassable road-blocks, nothing would *ever* change. Second, while we agree that faculty should not incur a significant increase in their responsibilities, we should ask the question "Can we switch to the radical alternative without a significant increase in faculty/staff workload?" We believe that the answer is "yes," for a few reasons:

²If you worry that students *won't* be motivated to do the work at all, see the 'Potential concerns' section below.

³If you worry that students *won't* engage in these practices, see the 'Potential concerns' section below.

- (a) We are willing to contribute a significant amount to the change effort.
- (b) The change can happen slowly, over the span of months or years, so that the regular effort required by faculty is small enough.
- (c) The change can happen on a small scale at first and increase in scope over time. For example, we could implement the radical alternative just for MTH 256 at first, leaving the remaining courses for later.

Students won't like change

Another downside to embracing the radical alternative is that students probably won't like adapting to the new system. We can see a few good reasons why students would resist this change:

- (a) Many students dislike exams, and for good reason.⁴ So, if the exam structure were to remain unchanged, students' resistance would be understandable.
- (b) Many students use homework grades to guide their studying. These students have adopting the learning process of completing whatever assignment is due next without asking questions like "How should I spend my time to best learn the material?" If we do not provide study guidance to students when adopting the radical alternative, these students would rightfully feel upset about losing guidance or support in their learning journey.

As you can tell, these reasons for resisting change are all dependent on *how* the radical alternative is adopted. We can prevent such frustrations if we consider our options for revising the exam structure and provide guidance on what students should study in order to be successful.

That said, these preventative measures will not work 100% of the time. Even if we recommend certain study habits or activities, some students will feel unguided, and no form of alternative assessment will ease all of the students' concerns. And, of course, some students will resist change no matter what the change is. Again, this seems like an unavoidable downside to adopting the radical alternative.

Potential concerns

The previous section covered unavoidable downsides to adopting the radical alternative. In this section, we examine some other potential concerns which we believe may have solutions.

Change won't happen

⁴See the "Exams are bad" section below.

A common reaction to systems-level change is “Maybe you’re right, but there’s no way that will ever happen!” This response is understandable because change takes time and money that the University would rather save than spend. But we should remember two important facts:

- (a) Whether change is possible misses the point. As educators, our goal is to achieve student learning in the best way possible. If there is a system that would serve students *way better* than the current one, that’s valuable information! And the likelihood of change is not exactly relevant to that discussion.
- (b) Change becomes possible if there is consensus. If the graduate students and instructors believe that change would be worthwhile, we might just be able to convince some of the administrators.

Students won’t practice or play

You may still think that, if we get rid of homework grades, students won’t engage with the material. Because of points we’ve already discussed, we think that this ultimately falls flat as a reason against switching to the radical alternative. Here’s a brief recap of those points:

- (a) Students already aren’t engaging. We agree that the number of homework submissions would likely drop if we were to adopt the radical alternative. But, as discussed in the section on the current system, we have evidence that many students are using AI platforms like Chat-GPT or Mathway to complete their homework assignments, and copying answers from other students has been an issue for years. In other words, the number of homework submissions is not necessarily indicative of how much students are practicing or playing with the content; a 20% decrease in homework submissions would not mean a 20% decrease in student engagement.
- (b) Intrinsic motivation > extrinsic motivation. We should re-emphasize that psychological research indicates that intrinsic motivators, such as learning the material and performing well on assessments, are more powerful than extrinsic motivators such as points or check-marks. So, this further casts doubt on the claim that student engagement would significantly decrease under the radical alternative.

Exams are bad, and homework points are cushions

The current system has students complete 5 written exams, one every other week, which contain at least 8 problems. This written exam structure has some obvious downsides, mainly that the exams are stressful and sometimes even unfair. This is often a reason teachers like homework points: they “cushion” the student’s grade, reducing the impact of high-stakes, unreasonable exams. If we adopt the radical alternative and keep the exam structure the same,

we would worsen these issues: exams would become even more high-stakes, only amplifying the stress for students. To make matters worse, abolishing homework grades would get rid of these cushion points, giving students no hope for making up for bad exam performance.

Importantly, these concerns arise *if* we keep the exam structure the same. In the 'Grading and assessment' section below, we look at some ways of getting around these downsides. In particular, we consider alternative forms of assessment and alternative grading scales that would make assessments low-stakes and fair.

Compromises

If neither the radical alternative nor the current system seem optimal, you may wonder if there are any good compromises between the two. Let's look at a few ways we might try to split the difference.

The reduction compromise

The *reduction compromise* is a system that does not eliminate homework grades entirely, but rather reduces their impact on the final grade. For example, instead of homework constituting 40% of a student's final grade, it could constitute only 20%. Another version of the reduction compromise would be to reduce the number of assignments, such as removing the online homework requirement or reducing the number of written homework assignments from 10 to 5.

Pros of the reduction compromise

Here's a brief list of reasons we might prefer the reduction compromise to the other options:

- (a) Practice is still required. If you aren't convinced that extrinsically motivating students to do homework is a bad thing, and if you worry that students will think "I am only doing the assignments that are graded" no matter what, then keeping homework as part of the grade may be a plus. The reduction compromise lets us do this while still reducing the amount of work required of the students.
- (b) Reduced workload gives students more agency. As in the radical alternative, the reduction compromise reduces the student workload. This would allow students to spend more time on problems which they find particularly difficult or interesting.
- (c) Reducing the contribution of homework is more likely to happen. We said that it is not a good idea to dwell on the question of "Will the administration agree to change?" right now. But the reduction compromise may ease your mind if you are worried that the department or university would never switch to the radical alternative.

- (d) Students will be familiar with the grading system. One of the cons of the radical alternative is that students won't like change. Since the reduction compromise doesn't get rid of homework entirely but instead merely shifts the grade percentages around, it may help in that regard.
- (e) We can pick-and-choose which assignments to cut. You may be sympathetic to the concerns we raised against the current system but still think that certain assignments are a necessary part of the grade. In this case, the reduction compromise would let us get rid of the unhelpful assignments while keeping the helpful ones.

Cons of the reduction compromise

Here's a brief list of problems or potential concerns with the reduction compromise:

- (a) Reducing extrinsic motivation may be confusing. We argued that one of the benefits of the radical alternative is that it removes extrinsic motivation from the course and encourages intrinsic motivation. But because the reduction compromise does not remove extrinsic motivation entirely, it is likely that a significant portion of students would still think "I am only doing the assignments that are graded." In our view, this would cause students to practice *less* than the current system or the radical alternative.
- (b) Good mistakes still get punished. One of the main benefits of the radical alternative is that mistakes during practice do not negatively impact the student's final grade. If we reduce the contribution of homework scores to the final grade, we can get closer to this ideal case, but at the end of the day students would still get punished for useful mistakes.
- (c) Students may still see school as work. In the first section, we discussed how the student workload leads students to view school as work instead of as a learning experience. While the radical alternative directly prevents this from happening by eliminating the student's workload entirely, the reduction compromise does not. So, some students will likely still focus on doing their work instead of focusing on learning. These students would still likely engage in cheating or dishonest behavior, and their appreciation, autonomy, and learning skills would suffer.

The student choice compromise

The *student choice compromise* is a system that allows students to choose how homework is used in the final grade. We'll look at the pros and cons of simple version of such a system, in which students pick between the current system and the radical alternative at the beginning of the term.

Pros of the student choice compromise

Here are some pros with the student choice compromise:

- (a) Giving students agency reduces student-centered concerns. One con of the radical alternative is that students won't like change. The student choice compromise would eliminate this issue, as any students afraid of change could opt to stay with the current system.
- (b) Problems with the radical alternative are less impactful. Likely, some students would opt to stay with the current system. This would mean that the downsides of the radical alternative would be less impactful than switching to the radical alternative outright. This may be especially desirable as a way to test the radical alternative before implementing it further.

Cons of the student choice compromise

Finally, here are some of the student choice compromise's cons:

- (a) Students won't know what's best. Students would probably have to choose between the two systems at the beginning of the term. But at the beginning of the term, students may not know what choice is best for them, and this would be especially true if this was the student's first math class at OSU. Additionally, the pros and cons of the two systems are not exactly obvious and students may not realize why something would be a pro or a con.
- (b) Students are a conflict of interest. We evaluated the different systems in terms of what would be best for student learning. But some students, especially those who adopt a school-as-work approach and want to get through their classes with as little effort as possible, won't agree that learning is the priority. Instead, they'll likely choose the current system because it rewards cheating or the use of AI.

Grading and assessment

Assessment is one of the most challenging parts of course design. How do we accurately assess student learning? What do we want students to leave the class understanding? Since these questions are both crucial and difficult, we will discuss a few ideas we have for grading and assessment in the current and radical systems.

Alternative summative assessment

Quizzes

A main concern of the radical alternative is that it emphasizes exam grades, which seems like a bad idea given how stressful (and sometimes unfair) written exams are. One idea for a different method of assessment is giving students weekly 15-30 minute quizzes. If we adopt such a quiz system along with the radical alternative, this would significantly reduce student stress as compared to the current exam structure. Additionally, a quiz structure would allow flexibility in the assessment content: half of the quizzes could focus on the behavioristic concepts we expect students to understand (using the power, quotient, and chain rule, etc.) while the other half of exams could focus on students applying these ideas to real world problems (solving word problems, conceptualizing results, etc.).

Take exam problems from homework

Another way of reducing stress and ensuring fair assessment comes from how we select exam problems. If we adopt the radical alternative, we could have the exam questions come straight from the ungraded homework assignments. This would enforce the idea that students should complete homework assignments and practice problems even though it is not required for their grade, further encouraging students to take charge of their own learning. This system would also provide students with clear expectations and a clear purpose behind the ungraded homework assignments.

Corrections

A final way we might fix the assessment structure is by allowing students to correct their exams or quizzes. For example, we could allow students to make up 20% of their lost assessment points by writing corrections for the problems they missed. Currently, exams are high-stakes and punishing for students whose initial level of understanding is low. If students had an opportunity to correct their mistakes, this would lower the stakes of assessment and force students to grow and learn from their mistakes. This also motivates students to fix early misconceptions and emphasizes understanding the material rather than memorizing it.

Grading

Averaging

As mathematicians, we know that taking the mean does not always accurately represent a given set of data. For example, consider the following exam grades: 91%, 92%, 40%, 94%, and 94%. The mean of these exams is a 82.2% while the median is a 92% and the mode is a 94%. In the current system, the mean represents the student's level of understanding. But which one best describes our student, who consistently performed at a high level with one horrible-no-good-very-bad day? Does it make sense to assign the student a 82.2% when they never completed an assignment at the B level? The median and mode are not only reasonable alternatives to averaging, but they are also more mathematically accurate than the average when there are outliers in a student's performance.

Minimum grading and the 0-4 scale

The typical 0 – 100% grading scale is disproportionately weighted towards failure, displaying that success is less likely than failure. Why would we delegate 60% to failure while only letting 20% of the grade display an appropriate level of learning (a B or higher)? This large scale results in inconsistent grading, it indicates that all points awarded during a given assignment hold the same weight, and it is generally mathematically inaccurate. Adding a minimum grade that students can earn in a class would allow for a more proportionate scale where the failure range and the passing range are closer in size. In [2], Feldman quotes researchers from the University of Massachusetts as saying

Minimum grading actually empowers teachers and schools rather than disempowers them, as it lessens, dampens out and neutralizes most of the negative aspects of grades and grading in school learning and in the behavioral processes while creating a climate of caring, hope and support, particularly for those students whose growth and development will most probably always be an intermittent and somewhat chaotic process and path. Minimum grading is a first step and key component in creating a culture of compassion and caring in a learning organization, classroom or school. (93)

Another approach to accurate grading is the 0-4 scale which has 5 levels of learning: 0, 1, 2, 3, and 4. On this scale, a 0 and a 1 are considered a failing grade while a 2, 3, or 4 are considered a passing grade. 0-4 grading solves the inconsistencies of the 0 – 100% scale and provides students with clearer expectations of what is needed to succeed in the course.

These approaches overall increase student motivation throughout the term even when they receive low scores on the first assignments (81). This is because it is mathematically possible for them to still pass the class and receive a grade appropriate to their level of learning, while the 0 – 100% scale makes this incredibly challenging. Additionally, while one might worry that minimum grading or the 0-4 scale would lead to grade inflation and a massive increase in passing rates, research shows that this does not happen in fact (92).

Conclusion

We and other members of the department believe that students in MTH 112Z or MTH 251-256 are required to complete too much work. Fortunately, we believe that there are realistic options for change that would significantly improve the learning experience of OSU's undergraduate math students. These options include a radical change of abolishing homework grades entirely, as well as several promising compromises. If these systems were paired with changes to assessment and the grading scale, students' experiences in calculus might involve more learning and playing than cheating and suffering.

References

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